DSA REFERENCE

2D Dynamic Programming

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Represent progress along two dimensions—positions, capacities, or strings—and reuse already solved cells.

VERSION 1.0 REVIEW
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STATUS HUMAN REVIEW REQUIRED

DSA / DYNAMIC-PROGRAMMING-2D

DSA

2D Dynamic Programming

CORE IDEA

Represent progress along two dimensions—positions, capacities, or strings—and reuse already solved cells.

Recognition cues

+ two sequences
+ grid paths
+ capacity plus item index
+ edit or alignment operations

Implementation sequence

+ name both axes
+ derive predecessor cells
+ initialize first row and column
+ verify traversal order
+ separate reconstruction from score

Invariant

Each cell answers the same question for a precisely smaller pair of dimensions.

Worked reasoning

Longest common subsequence compares final characters; equal characters extend the diagonal, otherwise take the better neighbor.

Complexity

Commonly O(rows × columns) time and space; rolling rows can reduce memory when reconstruction is unnecessary.

Failure modes

- [ ] off-by-one indexing between strings and table
- [ ] uninitialized boundaries
- [ ] reading future cells
- [ ] compressing before validating dependencies

Practice ladder

2D Dynamic Programming: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Unique Paths**	State the cue, invariant, edge case, and proof before implementation.
2	**Longest Common Subsequence**	State the cue, invariant, edge case, and proof before implementation.
3	**Edit Distance**	State the cue, invariant, edge case, and proof before implementation.
4	**Coin Change II**	State the cue, invariant, edge case, and proof before implementation.
5	**Distinct Subsequences**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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